

Integrals are also employed by engineers and scientists to evaluate the total amount or quantity of a given physical variable. The integral may be evaluated over a line, an area, or a volume. For example, the total mass of chemical contained in a reactor is given as the product of the concentration of chemical and the reactor volume, or

$$\text{Mass} = \text{concentration} \times \text{volume}$$

where concentration has units of mass per volume. However, suppose that concentration varies from location to location within the reactor. In this case, it is necessary to sum the products of local concentrations c_i and corresponding elemental volumes ΔV_i :

$$\text{Mass} = \sum_{i=1}^n c_i \Delta V_i$$

where n is the number of discrete volumes. For the continuous case, where $c(x, y, z)$ is a known function and x, y , and z are independent variables designating position in Cartesian coordinates, integration can be used for the same purpose:

$$\text{Mass} = \iiint c(x, y, z) dx dy dz$$

or

$$\text{Mass} = \iiint_V c(V) dV$$

which is referred to as a *volume integral*. Notice the strong analogy between summation and integration.

Similar examples could be given in other fields of engineering and science. For example, the total rate of energy transfer across a plane where the flux (in calories per square centimeter per second) is a function of position is given by

$$\text{Flux} = \iint_A u_x dA$$

which is referred to as an *areal integral*, where A = area.

These are just a few of the applications of integration that you might face regularly in the pursuit of your profession. When the functions to be analyzed are simple, you will normally choose to evaluate them analytically. However, it is often difficult or impossible when the function is complicated, as is typically the case in more realistic examples. In addition, the underlying function is often unknown and defined only by measurement at discrete points. For both these cases, you must have the ability to obtain approximate values for integrals using numerical techniques as described next.

19.2 NEWTON-COTES FORMULAS

The *Newton-Cotes formulas* are the most common numerical integration schemes. They are based on the strategy of replacing a complicated function or tabulated data with a polynomial that is easy to integrate:

$$I = \int_a^b f(x) dx \cong \int_a^b f_n(x) dx \quad (19.8)$$

where $f_n(x)$ = a polynomial of the form

$$f_n(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + a_nx^n \quad (19.9)$$

where n is the order of the polynomial. For example, in Fig. 19.4a, a first-order polynomial (a straight line) is used as an approximation. In Fig. 19.4b, a parabola is employed for the same purpose.

The integral can also be approximated using a series of polynomials applied piecewise to the function or data over segments of constant length. For example, in Fig. 19.5, three

FIGURE 19.4

The approximation of an integral by the area under (a) a straight line and (b) a parabola.

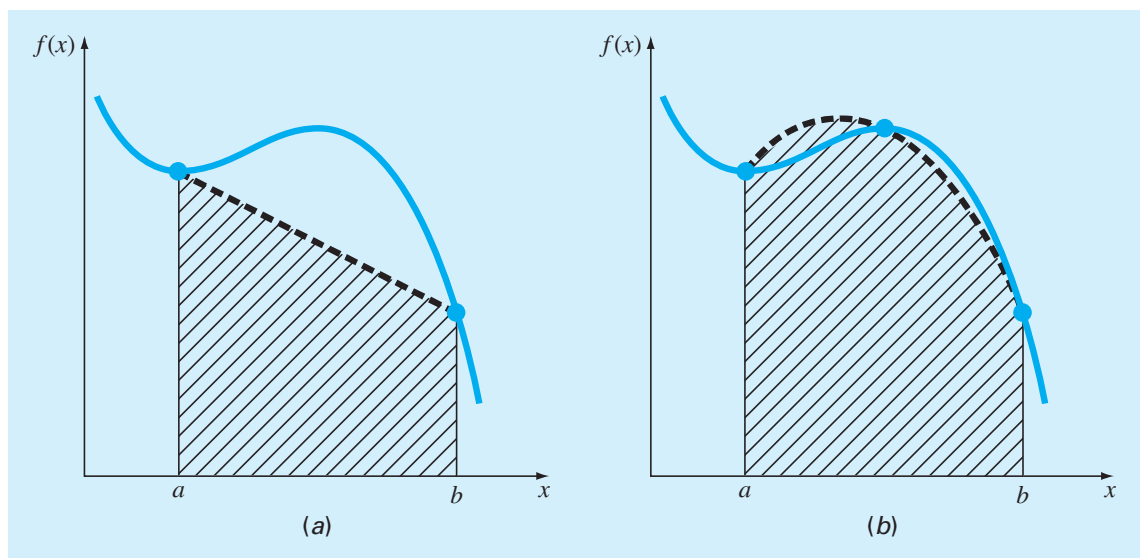
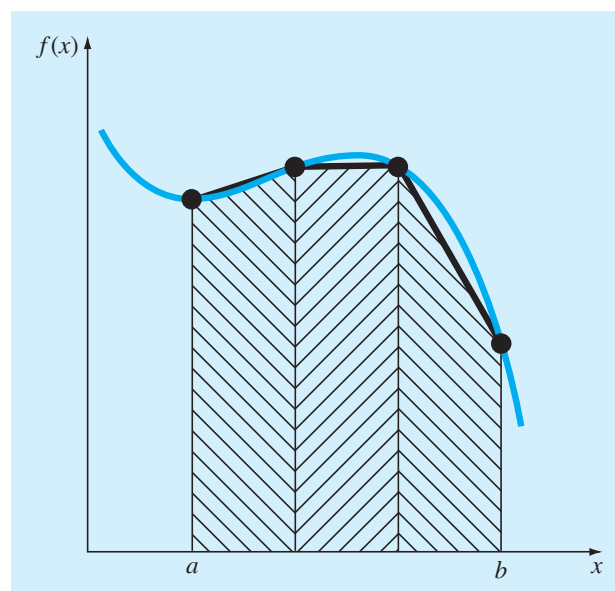
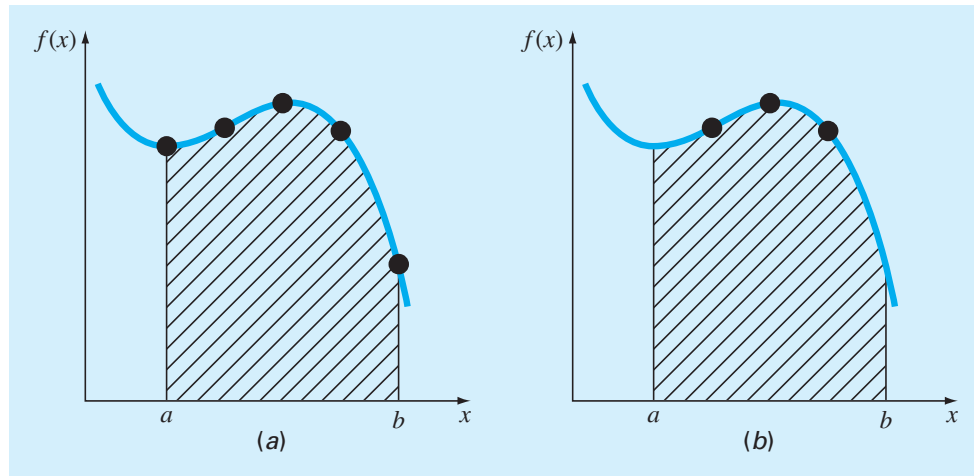


FIGURE 19.5

The approximation of an integral by the area under three straight-line segments.



**FIGURE 19.6**

The difference between (a) closed and (b) open integration formulas.

straight-line segments are used to approximate the integral. Higher-order polynomials can be utilized for the same purpose.

Closed and open forms of the Newton-Cotes formulas are available. The *closed forms* are those where the data points at the beginning and end of the limits of integration are known (Fig. 19.6a). The *open forms* have integration limits that extend beyond the range of the data (Fig. 19.6b). This chapter emphasizes the closed forms. However, material on open Newton-Cotes formulas is briefly introduced in Section 19.7.

19.3 THE TRAPEZOIDAL RULE

The *trapezoidal rule* is the first of the Newton-Cotes closed integration formulas. It corresponds to the case where the polynomial in Eq. (19.8) is first-order:

$$I = \int_a^b \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right] dx \quad (19.10)$$

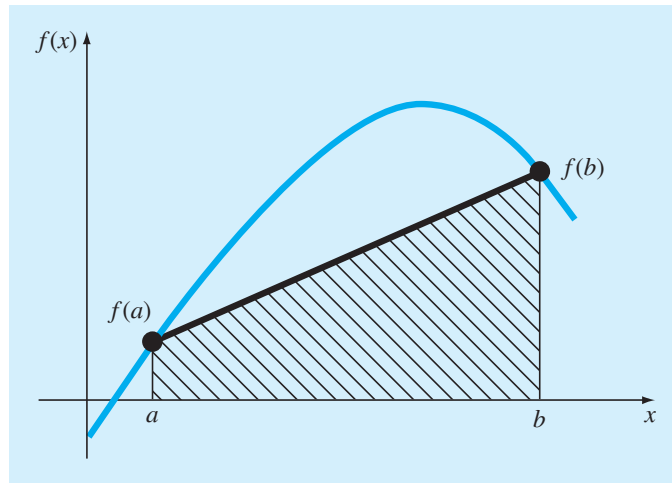
The result of the integration is

$$I = (b - a) \frac{f(a) + f(b)}{2} \quad (19.11)$$

which is called the *trapezoidal rule*.

Geometrically, the trapezoidal rule is equivalent to approximating the area of the trapezoid under the straight line connecting $f(a)$ and $f(b)$ in Fig. 19.7. Recall from geometry that the formula for computing the area of a trapezoid is the height times the average of the bases. In our case, the concept is the same but the trapezoid is on its side. Therefore, the integral estimate can be represented as

$$I = \text{width} \times \text{average height} \quad (19.12)$$

**FIGURE 19.7**

Graphical depiction of the trapezoidal rule.

or

$$I = (b - a) \times \text{average height} \quad (19.13)$$

where, for the trapezoidal rule, the average height is the average of the function values at the end points, or $[f(a) + f(b)]/2$.

All the Newton-Cotes closed formulas can be expressed in the general format of Eq. (19.13). That is, they differ only with respect to the formulation of the average height.

19.3.1 Error of the Trapezoidal Rule

When we employ the integral under a straight-line segment to approximate the integral under a curve, we obviously can incur an error that may be substantial (Fig. 19.8). An estimate for the local truncation error of a single application of the trapezoidal rule is

$$E_t = -\frac{1}{12} f''(\xi)(b - a)^3 \quad (19.14)$$

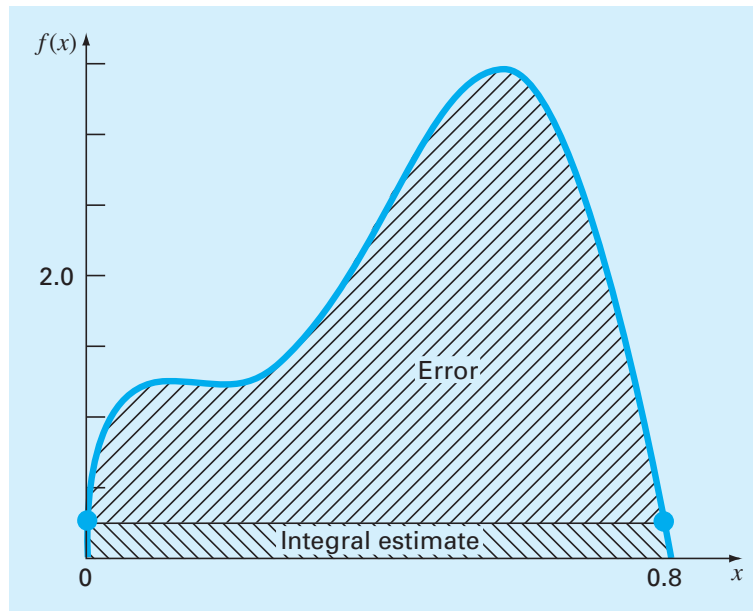
where ξ lies somewhere in the interval from a to b . Equation (19.14) indicates that if the function being integrated is linear, the trapezoidal rule will be exact because the second derivative of a straight line is zero. Otherwise, for functions with second- and higher-order derivatives (i.e., with curvature), some error can occur.

EXAMPLE 19.1 Single Application of the Trapezoidal Rule

Problem Statement. Use Eq. (19.11) to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$. Note that the exact value of the integral can be determined analytically to be 1.640533.

**FIGURE 19.8**

Graphical depiction of the use of a single application of the trapezoidal rule to approximate the integral of $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $x = 0$ to 0.8 .

Solution. The function values $f(0) = 0.2$ and $f(0.8) = 0.232$ can be substituted into Eq. (19.11) to yield

$$I = (0.8 - 0) \frac{0.2 + 0.232}{2} = 0.1728$$

which represents an error of $E_t = 1.640533 - 0.1728 = 1.467733$, which corresponds to a percent relative error of $\varepsilon_t = 89.5\%$. The reason for this large error is evident from the graphical depiction in Fig. 19.8. Notice that the area under the straight line neglects a significant portion of the integral lying above the line.

In actual situations, we would have no foreknowledge of the true value. Therefore, an approximate error estimate is required. To obtain this estimate, the function's second derivative over the interval can be computed by differentiating the original function twice to give

$$f''(x) = -400 + 4,050x - 10,800x^2 + 8,000x^3$$

The average value of the second derivative can be computed as [Eq. (19.7)]

$$\bar{f}''(x) = \frac{\int_0^{0.8} (-400 + 4,050x - 10,800x^2 + 8,000x^3) dx}{0.8 - 0} = -60$$

which can be substituted into Eq. (19.14) to yield

$$E_a = -\frac{1}{12}(-60)(0.8)^3 = 2.56$$

which is of the same order of magnitude and sign as the true error. A discrepancy does exist, however, because of the fact that for an interval of this size, the average second derivative is not necessarily an accurate approximation of $f''(\xi)$. Thus, we denote that the error is approximate by using the notation E_a , rather than exact by using E_t .

19.3.2 The Composite Trapezoidal Rule

One way to improve the accuracy of the trapezoidal rule is to divide the integration interval from a to b into a number of segments and apply the method to each segment (Fig. 19.9). The areas of individual segments can then be added to yield the integral for the entire interval. The resulting equations are called *composite*, or *multiple-segment*, *integration formulas*.

Figure 19.9 shows the general format and nomenclature we will use to characterize composite integrals. There are $n + 1$ equally spaced base points ($x_0, x_1, x_2, \dots, x_n$). Consequently, there are n segments of equal width:

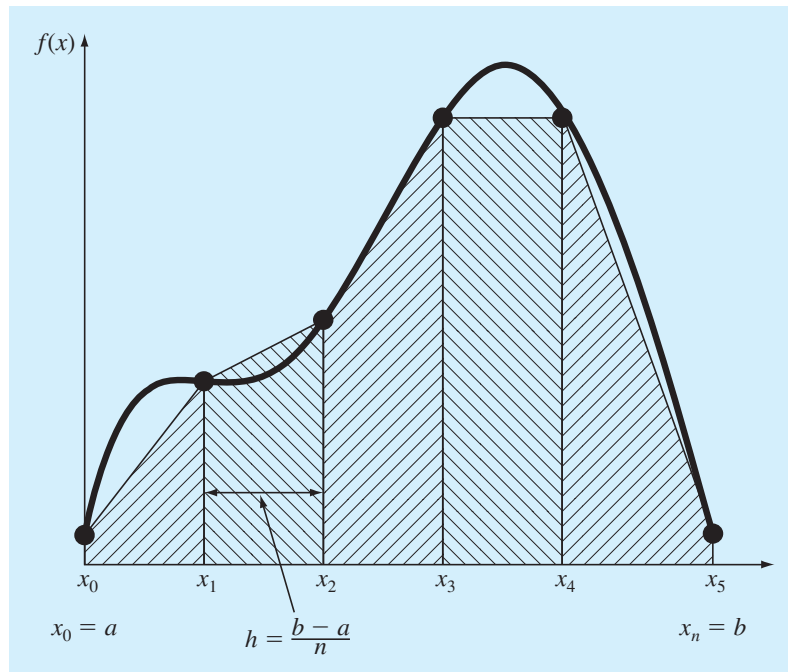
$$h = \frac{b - a}{n} \quad (19.15)$$

If a and b are designated as x_0 and x_n , respectively, the total integral can be represented as

$$I = \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \cdots + \int_{x_{n-1}}^{x_n} f(x) dx$$

FIGURE 19.9

Composite trapezoidal rule.



Substituting the trapezoidal rule for each integral yields

$$I = h \frac{f(x_0) + f(x_1)}{2} + h \frac{f(x_1) + f(x_2)}{2} + \cdots + h \frac{f(x_{n-1}) + f(x_n)}{2} \quad (19.16)$$

or, grouping terms:

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right] \quad (19.17)$$

or, using Eq. (19.15) to express Eq. (19.17) in the general form of Eq. (19.13):

$$I = \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}} \quad (19.18)$$

Because the summation of the coefficients of $f(x)$ in the numerator divided by $2n$ is equal to 1, the average height represents a weighted average of the function values. According to Eq. (19.18), the interior points are given twice the weight of the two end points $f(x_0)$ and $f(x_n)$.

An error for the composite trapezoidal rule can be obtained by summing the individual errors for each segment to give

$$E_t = -\frac{(b-a)^3}{12n^3} \sum_{i=1}^n f''(\xi_i) \quad (19.19)$$

where $f''(\xi_i)$ is the second derivative at a point ξ_i located in segment i . This result can be simplified by estimating the mean or average value of the second derivative for the entire interval as

$$\bar{f}'' \cong \frac{\sum_{i=1}^n f''(\xi_i)}{n} \quad (19.20)$$

Therefore $\sum f''(\xi_i) \cong n \bar{f}''$ and Eq. (19.19) can be rewritten as

$$E_a = -\frac{(b-a)^3}{12n^2} \bar{f}'' \quad (19.21)$$

Thus, if the number of segments is doubled, the truncation error will be quartered. Note that Eq. (19.21) is an approximate error because of the approximate nature of Eq. (19.20).

EXAMPLE 19.2 Composite Application of the Trapezoidal Rule

Problem Statement. Use the two-segment trapezoidal rule to estimate the integral of

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

from $a = 0$ to $b = 0.8$. Employ Eq. (19.21) to estimate the error. Recall that the exact value of the integral is 1.640533.

Solution. For $n = 2$ ($h = 0.4$):

$$f(0) = 0.2 \quad f(0.4) = 2.456 \quad f(0.8) = 0.232$$

$$I = 0.8 \frac{0.2 + 2(2.456) + 0.232}{4} = 1.0688$$

$$E_t = 1.640533 - 1.0688 = 0.57173 \quad \varepsilon_t = 34.9\%$$

$$E_a = -\frac{0.8^3}{12(2)^2}(-60) = 0.64$$

where -60 is the average second derivative determined previously in Example 19.1.

The results of the previous example, along with three- through ten-segment applications of the trapezoidal rule, are summarized in Table 19.1. Notice how the error decreases as the number of segments increases. However, also notice that the rate of decrease is gradual. This is because the error is inversely related to the square of n [Eq. (19.21)]. Therefore, doubling the number of segments quarters the error. In subsequent sections we develop higher-order formulas that are more accurate and that converge more quickly on the true integral as the segments are increased. However, before investigating these formulas, we will first discuss how MATLAB can be used to implement the trapezoidal rule.

19.3.3 MATLAB M-file: trap

A simple algorithm to implement the composite trapezoidal rule can be written as in Fig. 19.10. The function to be integrated is passed into the M-file along with the limits of integration and the number of segments. A loop is then employed to generate the integral following Eq. (19.18).

TABLE 19.1 Results for the composite trapezoidal rule to estimate the integral of $f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$ from $x = 0$ to 0.8 . The exact value is 1.640533.

n	h	I	ε_t (%)
2	0.4	1.0688	34.9
3	0.2667	1.3695	16.5
4	0.2	1.4848	9.5
5	0.16	1.5399	6.1
6	0.1333	1.5703	4.3
7	0.1143	1.5887	3.2
8	0.1	1.6008	2.4
9	0.0889	1.6091	1.9
10	0.08	1.6150	1.6